

A Survey of Bayesian Deep Learning for Uncertainty Quantification

InteractiveSurvey

Abstract

The rapid deployment of deep neural networks in safety-critical domains has necessitated a paradigm shift from purely accuracy-driven models to those capable of reliable uncertainty quantification. While deterministic architectures often exhibit unwarranted overconfidence on out-of-distribution data, Bayesian Deep Learning offers a rigorous framework for distinguishing between reducible model ignorance and irreducible data noise. This survey provides a comprehensive examination of Bayesian methodologies specifically designed for uncertainty quantification, synthesizing recent advances in stochastic inference, calibration mechanisms, and uncertainty decomposition. Unlike previous reviews focusing narrowly on algorithms or applications, this work critically analyzes the interplay between theoretical fidelity and computational practicality across diverse Bayesian approximations. We establish that while sampling-based approaches offer theoretical guarantees, optimization-based variational strategies provide essential scalability, though often at the cost of posterior expressiveness. Furthermore, the analysis reveals that intrinsic calibration via proper scoring rules fosters naturally well-calibrated models, whereas post-hoc techniques, while efficient, frequently fail under dynamic distribution shifts. By bridging abstract probabilistic theory with empirical performance metrics, this paper establishes a unified perspective on constructing trustworthy deep learning systems that accurately articulate their own limitations.

1 Introduction

The rapid deployment of deep neural networks in safety-critical domains, ranging from autonomous driving to medical diagnostics, has necessitated a paradigm shift from purely accuracy-driven models to those capable of reliable uncertainty quan-

tification. While deterministic architectures excel at pattern recognition within known distributions, they frequently exhibit unwarranted overconfidence when encountering out-of-distribution data or ambiguous inputs, posing significant risks in high-stakes decision-making scenarios. Consequently, the ability to distinguish between reducible model ignorance and irreducible data noise has become a fundamental requirement for robust artificial intelligence. Bayesian Deep Learning emerges as a rigorous theoretical framework to address these challenges by treating network weights as probability distributions rather than fixed point estimates, thereby enabling the propagation of uncertainty through the inference process and providing statistically grounded confidence measures alongside predictions [1].

This survey provides a comprehensive examination of Bayesian Deep Learning methodologies specifically designed for uncertainty quantification, synthesizing recent advances in stochastic inference, calibration mechanisms, and uncertainty decomposition [1]. Unlike previous reviews that often focus narrowly on algorithmic implementation or specific application domains, this work critically analyzes the interplay between theoretical fidelity and computational practicality across diverse Bayesian approximations. We systematically evaluate how different inference strategies impact the quality of predictive distributions, with a particular emphasis on the trade-offs between asymptotic exactness and scalability in high-dimensional settings. By bridging the gap between abstract probabilistic theory and empirical performance metrics, this paper aims to establish a unified perspective on constructing trustworthy deep learning systems that can accurately articulate their own limitations.

The initial portion of this review establishes a robust empirical evaluation framework, scrutinizing the comparative efficacy of Markov Chain

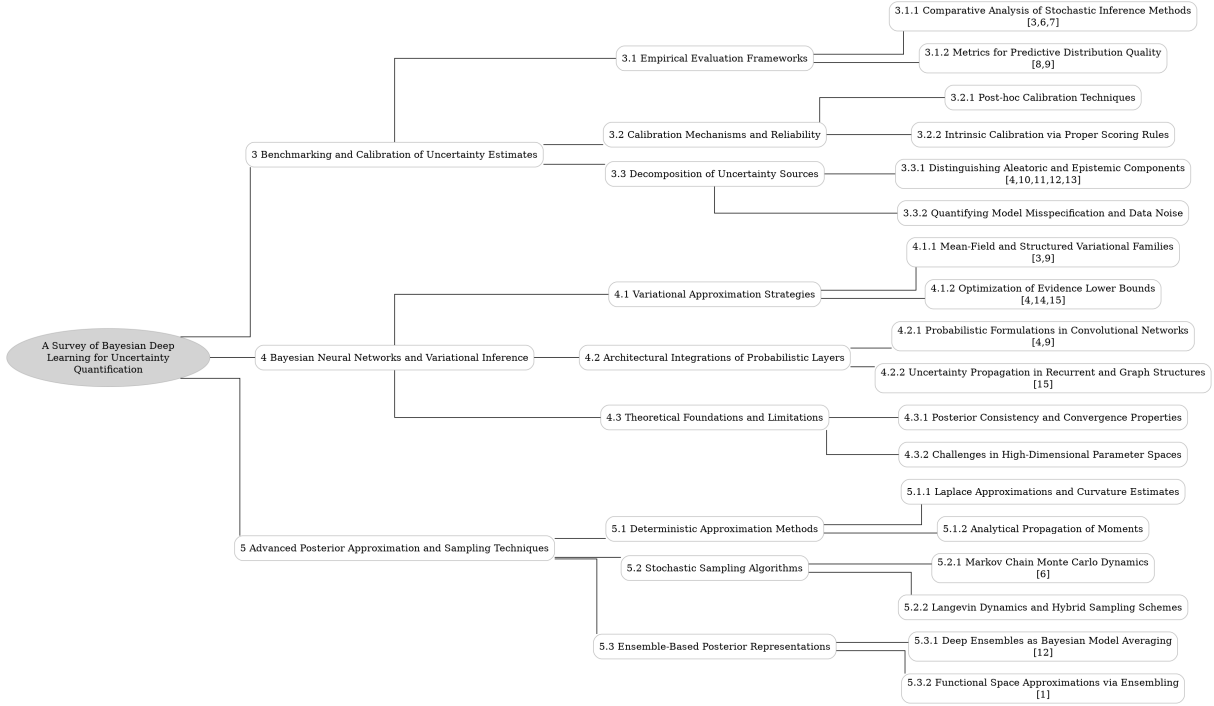


Figure 1: The outline of our survey: A Survey of Bayesian Deep Learning for Uncertainty Quantification

$$\hat{q}(\theta) = \underset{q(\theta) \in \mathcal{Q}}{\operatorname{argmin}} d_{\text{KL}}(q(\theta), p(\theta|\mathcal{D}))$$

Figure 2: Chart from 'LEARNING ACTIVE SUBSPACES FOR EFFECTIVE AND SCALABLE UNCERTAINTY QUANTIFICATION IN DEEP NEURAL NETWORKS' (arXiv:2309.03061)

Monte Carlo and Variational Inference techniques. We analyze how stochastic inference methods diverge in their handling of posterior complexity, noting that while sampling-based approaches offer theoretical guarantees, optimization-based variational strategies provide essential scalability for modern architectures. This discussion extends to a detailed assessment of metrics for predictive distribution quality, moving beyond standard accuracy scores to incorporate proper scoring rules, reliability diagrams, and coverage rates. These tools are essential for diagnosing miscalibration and ensuring that predicted confidence intervals genuinely reflect empirical frequencies, particularly under distribution shifts where deterministic models typically fail.

Subsequently, the survey investigates calibration mechanisms, contrasting post-hoc adjustment techniques with intrinsic calibration strategies driven by proper scoring rules. We explore how

methods like temperature scaling can correct systematic overconfidence in pre-trained models without retraining, while also highlighting their limitations in dynamic, non-stationary environments. In parallel, we examine intrinsic approaches that integrate calibration objectives directly into the learning process, fostering models that naturally output well-calibrated probabilities. This section further delves into the critical decomposition of uncertainty sources, distinguishing between aleatoric uncertainty arising from data noise and epistemic uncertainty stemming from model misspecification [2]. Understanding this distinction is vital for applications such as active learning and risk assessment, where the reducibility of uncertainty dictates the appropriate mitigation strategy.

Finally, the paper synthesizes these themes through an in-depth analysis of Bayesian Neural Networks and variational approximation strategies, focusing on the evolution from restrictive mean-field assumptions to more expressive structured variational families [3]. We discuss how advanced approximations, including normalizing flows and hierarchical priors, attempt to capture complex multi-modal posterior structures that simple Gaussian assumptions often miss. This comprehensive overview connects theoretical developments with practical benchmarking results, offering guidance on selecting appropriate modeling

$$z = \frac{x - \mu}{\sigma}$$

Figure 3: Chart from 'Bayesian Optimized Deep Ensemble for Uncertainty Quantification of Deep Neural Networks: a System Safety Case Study on Sodium Fast Reactor Thermal Stratification Modeling' (arXiv:2412.08776)

choices based on specific operational constraints and uncertainty requirements. The synthesis of these topics provides a holistic view of the current landscape, identifying key bottlenecks in existing methodologies and setting the stage for future research directions.

$$conf(\hat{\pi}) = \frac{1}{n} \sum_{i=1}^n \hat{\pi}_i.$$

Figure 4: Chart from 'Comparing the quality of neural network uncertainty estimates for classification problems' (arXiv:2308.05903)

The primary contributions of this survey are threefold. First, we provide a unified taxonomy of Bayesian uncertainty quantification methods that clarifies the relationships between inference algorithms, calibration techniques, and uncertainty decomposition strategies [4]. Second, we offer a critical comparative analysis of empirical evaluation frameworks, highlighting the strengths and weaknesses of various metrics in detecting miscalibration and distinguishing uncertainty sources under realistic conditions. Third, by synthesizing insights from both theoretical probability and applied deep learning, we identify persistent gaps in current research, particularly regarding the robustness of uncertainty estimates under severe model misspecification and distribution shifts, thereby outlining a clear roadmap for developing more reliable and interpretable Bayesian deep learning systems [5].

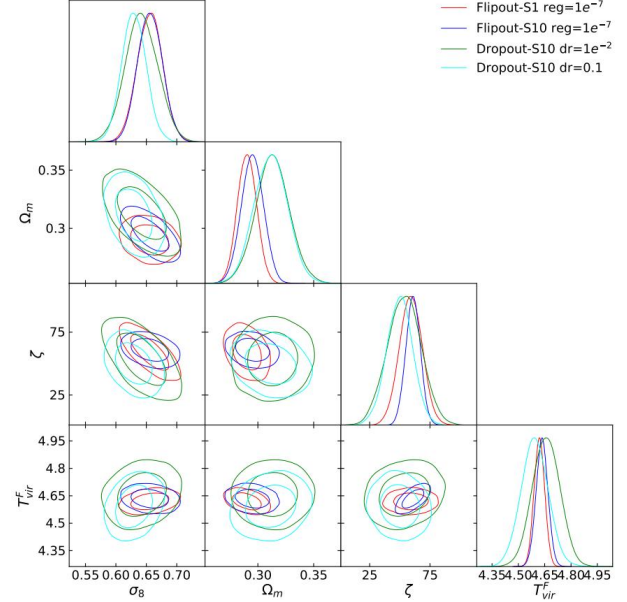


Figure 5: Chart from 'PARAMETERS ESTIMATION FROM THE 21 CM SIGNAL USING VARIATIONAL INFERENCE' (arXiv:2005.02299)

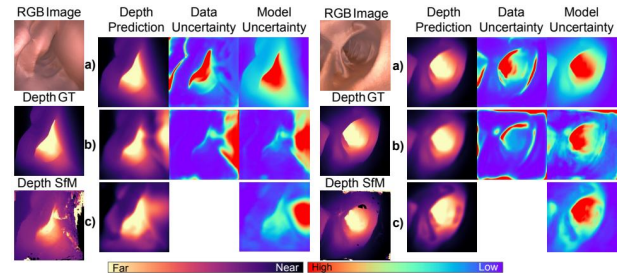


Figure 6: Chart from 'On the Uncertain Single-View Depths in Colonoscopies' (arXiv:2112.08906)

2 Benchmarking and Calibration of Uncertainty Estimates

2.1 Empirical Evaluation Frameworks

2.1.1 Comparative Analysis of Stochastic Inference Methods

Stochastic inference methods for Bayesian neural networks primarily diverge into Markov Chain Monte Carlo (MCMC) and Variational Inference (VI), each presenting distinct trade-offs between asymptotic accuracy and computational scalability [3]. MCMC techniques, such as Stochastic Gradient Hamiltonian Monte Carlo, offer theoretically exact posterior samples by constructing Markov chains that converge to the true distribution [6]. However, their application in high-dimensional deep learning contexts is often prohibitive due to slow mixing rates and substantial computational overhead, rendering them impractical for large-scale datasets or real-time inference requirements where rapid convergence is essential.

$$A(y', y^t | x) = \min \left(1, \underbrace{\frac{g(y^t | y', x)}{g(y' | y^t, x)}}_{\text{transitions}} \underbrace{\frac{p(x | y', \theta, \mathcal{D}, \mathcal{M})}{p(x | y^t, \theta, \mathcal{D}, \mathcal{M})}}_{\text{likelihoods}} \underbrace{\frac{p(y')}{p(y^t)}}_{\text{priors}} \right)$$

Figure 7: Chart from 'Quantification of Predictive Uncertainty via Inference-Time Sampling' (arXiv:2308.01731)

In contrast, VI approximates the intractable posterior with a tractable variational distribution by minimizing the Kullback-Leibler divergence, effectively maximizing the Evidence Lower Bound [7]. This approach transforms inference into an optimization problem, enabling efficient integration with standard gradient-based learning frameworks like stochastic gradient descent. While methods such as Bayes by Backpropagation and Monte Carlo Dropout significantly reduce computational costs and facilitate scalability, they frequently suffer from underestimation of posterior variance. This limitation arises from the restrictive assumptions imposed on the approximating family, such as mean-field Gaussian distributions, which may fail to capture complex multi-modal structures inherent in the true posterior landscape.

Recent comparative analyses highlight that the efficacy of these methods is heavily contingent on the interaction between the chosen inference algorithm and the expressiveness of the approximating family. Advanced variants, including normalizing flows and natural gradient updates, attempt

$$\hat{\Sigma}_{y_i} = \frac{1}{N_s - 1} \sum_{j=1}^{N_s} \epsilon_{j,i} \epsilon_{j,i}^\top,$$

Figure 8: Chart from 'Bayesian Learning for Neural Networks: an algorithmic survey' (arXiv:2211.11865)

to bridge the fidelity gap by employing more flexible distributions or optimizing the geometry of the parameter space. Nevertheless, empirical evaluations suggest that while VI-based approaches generally provide superior utility in terms of speed and ease of implementation, MCMC remains the gold standard for calibration when computational resources permit. Consequently, selecting an appropriate stochastic inference strategy requires a nuanced understanding of the specific uncertainty sources—epistemic versus aleatoric—and the operational constraints of the target application.

2.1.2 Metrics for Predictive Distribution Quality

Evaluating the quality of predictive distributions requires metrics that assess both calibration and sharpness, ensuring predicted probabilities align with empirical frequencies while maintaining informative confidence intervals. Proper scoring rules, such as the negative log-likelihood and Brier score, serve as foundational tools by penalizing overconfident incorrect predictions and rewarding well-calibrated uncertainty estimates. Unlike accuracy-focused metrics that only consider the mode of the distribution, these scores evaluate the entire probability density function, making them essential for identifying models that may appear accurate but suffer from poor uncertainty quantification, particularly in out-of-distribution scenarios where deterministic approaches often fail.

Beyond scalar scoring rules, reliability diagrams and Expected Calibration Error (ECE) provide granular insights into the agreement between predicted confidence and actual correctness across different probability bins [8]. A reliable model should exhibit higher uncertainty for misclassified samples compared to correct ones, a property often verified through separation metrics like the Area Under the Receiver Operating Characteristic curve between uncertainty scores and prediction errors. Furthermore, predictive entropy quantifies the average information content within the

distribution, offering a direct measure of epistemic and aleatoric uncertainty; however, relying solely on marginalized entropy can obscure multimodal structures, necessitating joint distribution analysis to fully capture complex predictive behaviors [9].

$$\text{CRPS} = \frac{1}{n} \sum_{i=1}^n \int_{-\infty}^{\infty} [F(x) - \mathbb{I}(x \geq y_i)]^2 dx,$$

Figure 9: Chart from 'From Aleatoric to Epistemic: Exploring Uncertainty Quantification Techniques in Artificial Intelligence' (arXiv:2501.03282)

$$\begin{aligned} I_{p(y^*|x^*, X, Y)}[y^*] \\ = - \sum_{y^* \in \{\text{malignant}, \text{benign}\}} p(y^* = c|x^*, X, Y) \log p(y^* = c) \end{aligned}$$

Figure 10: Chart from 'BI-RADS prediction of mammographic masses using uncertainty information extracted from a Bayesian Deep Learning model' (arXiv:2503.13999)

To address limitations in parametric assumptions, distribution-free frameworks such as Conformal Prediction offer rigorous guarantees on coverage probabilities without relying on specific distributional forms. These methods transform heuristic confidence scores into statistically valid prediction sets, ensuring that the ground truth falls within the predicted interval with a user-specified probability. While traditional metrics like mean squared error evaluate point estimates, comprehensive assessment of predictive distribution quality increasingly integrates coverage rates, interval widths, and calibration plots to validate that uncertainty estimates remain robust under distribution shifts and effectively distinguish between high-confidence accurate predictions and uncertain, potentially erroneous outputs.

2.2 Calibration Mechanisms and Reliability

2.2.1 Post-hoc Calibration Techniques

Post-hoc calibration techniques address the systematic overconfidence observed in modern deep neural networks by adjusting output probabilities after the initial training phase, utilizing a separate held-out calibration dataset. Unlike methods that integrate uncertainty estimation directly into the learning objective, these approaches treat the pre-trained model as a fixed feature extractor and learn a mapping function to transform raw logits into well-calibrated probabilities. This paradigm

is particularly advantageous for legacy systems or complex architectures where retraining is computationally prohibitive, allowing for the correction of miscalibration without altering the underlying network weights or architecture.

The most prevalent strategies include Platt scaling, which fits a logistic regression model to the network outputs, and temperature scaling, a simplified variant that learns a single scalar parameter to soften the probability distribution. While Platt scaling offers flexibility through linear transformation, temperature scaling has gained prominence for its simplicity and effectiveness in reducing Expected Calibration Error (ECE) with minimal risk of overfitting due to its single degree of freedom. These methods operate under the assumption that the model's ranking of predictions remains correct, seeking only to adjust the confidence magnitudes to align with empirical accuracy across different confidence bins, thereby restoring the diagonal relationship in reliability diagrams.

Despite their efficacy in independent and identically distributed settings, post-hoc methods face significant challenges when applied to out-of-distribution data or tasks involving complex dependency structures, such as pixel-wise segmentation. Since the calibration mapping is static and learned solely from the calibration set, it may fail to generalize under distribution shifts where the nature of the model's overconfidence changes dynamically. Furthermore, while these techniques effectively minimize scalar calibration metrics, they do not inherently improve the sharpness of predictive distributions or capture epistemic uncertainty, often necessitating hybrid approaches that combine post-hoc adjustment with inherent uncertainty modeling for robust performance in safety-critical applications.

2.2.2 Intrinsic Calibration via Proper Scoring Rules

Intrinsic calibration via proper scoring rules offers a rigorous framework for evaluating probabilistic forecasts without relying on post-hoc adjustments like temperature scaling. Unlike reliability diagrams that assess calibration through binning strategies, proper scoring rules, such as the Brier score and logarithmic loss, provide a strictly consistent evaluation metric that simultaneously penalizes both miscalibration and lack of sharpness. These metrics quantify the divergence between predicted probability distributions and ob-

served outcomes, ensuring that the model’s inherent uncertainty estimates align with empirical frequencies directly during the optimization phase.

The mathematical formulation of these scores encourages models to output probabilities that reflect true likelihoods, thereby minimizing the expected loss when the forecast matches the ground truth distribution. For instance, the Brier score computes the mean squared difference between predicted probabilities and binary outcomes, offering a differentiable objective that can be integrated directly into the training loop of deep neural networks. This approach mitigates the common tendency of modern architectures to produce overconfident predictions, as the penalty for assigning high probability to incorrect classes is severe, forcing the network to learn more representative uncertainty bounds intrinsically rather than correcting them externally.

Furthermore, leveraging proper scoring rules facilitates robust performance under distribution shifts, where traditional accuracy metrics often fail to capture degradation in predictive reliability. By optimizing for these strictly proper rules, models develop a nuanced understanding of epistemic and aleatoric uncertainty, leading to prediction intervals that maintain coverage even in out-of-distribution scenarios. This intrinsic alignment is particularly critical in safety-critical applications, as it ensures that the model’s confidence serves as a trustworthy indicator of potential error, eliminating the dependency on held-out calibration datasets that may not represent the operational environment.

2.3 Decomposition of Uncertainty Sources

2.3.1 Distinguishing Aleatoric and Epistemic Components

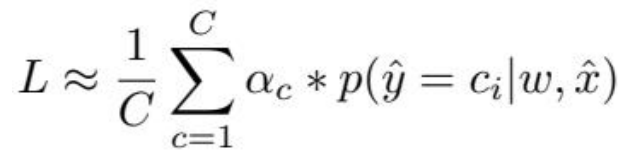
The fundamental distinction between aleatoric and epistemic uncertainty lies in their ontological origins and reducibility [10]. Aleatoric uncertainty, often termed statistical or irreducible uncertainty, arises from inherent noise within the data generation process, such as sensor measurement errors or stochastic environmental dynamics [11]. Conversely, epistemic uncertainty, also known as model or systematic uncertainty, stems from a lack of knowledge regarding the true underlying function, typically due to insufficient training data or model misspecification. While aleatoric uncertainty persists regardless of data volume,

epistemic uncertainty is theoretically reducible through the acquisition of additional informative samples or improved model architectures [10].



$$CV = \sqrt{\frac{\sum (V_i - \mu)^2}{N \cdot \mu^2}}.$$

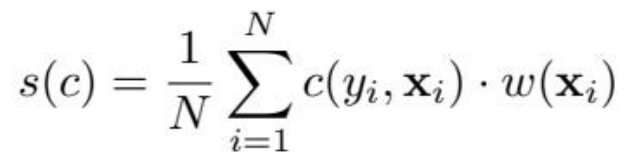
Figure 11: Chart from ‘Bayesian Neural Networks for Uncertainty Estimation of Imaging Biomarkers’ (arXiv:2008.12680)



$$L \approx \frac{1}{C} \sum_{c=1}^C \alpha_c * p(\hat{y} = c_i | w, \hat{x})$$

Figure 12: Chart from ‘Estimating Uncertainty and Interpretability in Deep Learning for Coronavirus (COVID-19) Detection’ (arXiv:2003.10769)

Operationally disentangling these components requires decomposing the total predictive uncertainty into variance attributable to data noise and variance arising from parameter ambiguity. In Bayesian frameworks, this is frequently achieved by analyzing the posterior predictive distribution, where total uncertainty is expressed as the sum of the expected variance (aleatoric) and the variance of the expectations (epistemic) [4]. Techniques such as Deep Ensembles and Monte Carlo Dropout approximate this decomposition by sampling from diverse model configurations; high disagreement among ensemble members indicates significant epistemic uncertainty, whereas consistent predictions with high residual error suggest dominant aleatoric factors [12].



$$s(c) = \frac{1}{N} \sum_{i=1}^N c(y_i, \mathbf{x}_i) \cdot w(\mathbf{x}_i)$$

Figure 13: Chart from ‘Constraint-Aware Neurosymbolic Uncertainty Quantification with Bayesian Deep Learning for Scientific Discovery’ (arXiv:20601.12442)

However, the separation of these uncertainties is not always absolute and can depend heavily on the chosen model class and feature representation.

Certain methodologies treat all uncertainty as a unified expression of rational belief, arguing that what appears as aleatoric in one modeling context may be reducible epistemic uncertainty in another with expanded feature spaces [13]. Furthermore, recent critiques highlight that standard decomposition metrics may conflate systematic errors with stochastic noise if the model structure is inadequate. Consequently, while distinguishing these components is vital for risk-sensitive decision-making and active learning strategies, the boundary remains context-dependent, necessitating careful interpretation of uncertainty estimates relative to the specific modeling assumptions employed.

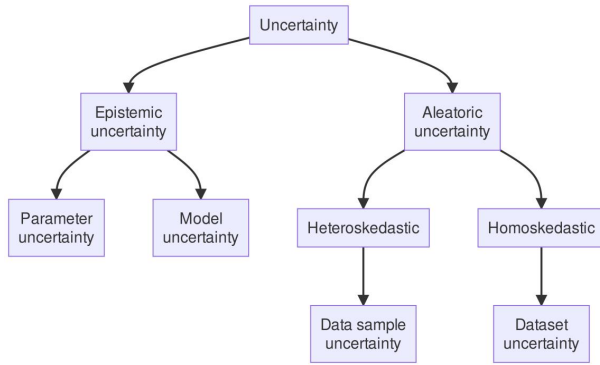


Figure 14: Chart from 'BORCH: A DEEP UNIVERSAL PROBABILISTIC PROGRAMMING LANGUAGE' (arXiv:2209.06168)

2.3.2 Quantifying Model Misspecification and Data Noise

Quantifying model misspecification involves assessing the discrepancy between the assumed hypothesis space and the true data-generating process, a primary source of epistemic uncertainty. When neural networks operate under incorrect architectural assumptions or insufficient capacity, they often yield overconfident predictions despite high error rates, particularly under distributional shifts. Traditional calibration metrics, such as Expected Calibration Error, frequently fail to distinguish between uncertainty arising from limited data and that stemming from fundamental structural biases, leading to unreliable confidence estimates in safety-critical applications where the model form itself is inadequate.

Concurrently, data noise, or aleatoric uncertainty, represents inherent stochasticity within the observation process that cannot be reduced by acquiring additional samples. This includes sensor

errors, labeling inconsistencies, and intrinsic variability in the underlying phenomena. Effective quantification requires decoupling this irreducible noise from model ignorance, often achieved by predicting distributional parameters rather than point estimates. However, when data noise is heteroscedastic or non-Gaussian, standard likelihood assumptions introduce further misspecification, complicating the isolation of pure data-driven uncertainty from artifacts introduced by inappropriate noise modeling choices.

Recent frameworks aim to jointly quantify these intertwined sources by leveraging Bayesian approximations and ensemble methods to capture predictive variance components separately. By injecting controlled noise during training and evaluating performance under out-of-distribution conditions, researchers can better attribute uncertainty to either data quality or model limitations. Nevertheless, current approaches often rely on convenient approximations like Monte Carlo dropout, which may not fully capture true model uncertainty under severe misspecification. Rigorous benchmarking protocols are therefore essential to validate whether reported uncertainty scores genuinely reflect the combined impact of structural model deficits and noisy observations.

3 Bayesian Neural Networks and Variational Inference

3.1 Variational Approximation Strategies

3.1.1 Mean-Field and Structured Variational Families

Mean-field variational inference (MFVI) represents the predominant approximation strategy in Bayesian neural networks, primarily due to its computational tractability in high-dimensional parameter spaces [3]. This approach posits that the variational posterior distribution factorizes completely over individual network weights, typically assuming independent Gaussian distributions for each parameter. By enforcing this strict independence assumption, MFVI simplifies the optimization of the evidence lower bound (ELBO) and enables efficient gradient-based learning through techniques like the reparameterization trick. However, this factorization inherently neglects posterior correlations between weights, often leading to underestimation of epistemic uncertainty and potentially degraded predictive performance in complex, multi-modal landscapes.

To address the limitations of fully factorized approximations, structured variational families have been developed to capture dependencies within subsets of parameters while maintaining computational feasibility. These methods relax the mean-field assumption by introducing block-diagonal covariance structures or employing normalizing flows that transform simple base distributions into more flexible, correlated posteriors. Structured approaches aim to balance the expressiveness required to model intricate weight interactions with the scalability necessary for deep architectures. Techniques such as multiplicative normalizing flows and low-rank approximations allow the variational distribution to better align with the true posterior geometry, thereby improving uncertainty quantification without incurring the prohibitive costs associated with full-covariance matrices.

Despite these advancements, the choice between mean-field and structured families involves a critical trade-off between approximation accuracy and optimization stability. While structured families theoretically offer superior fidelity to the true posterior, they often introduce additional hyperparameters and optimization challenges that can hinder convergence in very deep networks. Conversely, mean-field approximations remain robust and easy to implement, making them the default choice for many practical applications despite their known tendency to ignore data in wide network limits. Current research continues to explore hybrid strategies that selectively introduce structure where correlations are most significant, aiming to achieve a more accurate representation of uncertainty without sacrificing the scalability that makes variational inference attractive for large-scale deep learning [9].

3.1.2 Optimization of Evidence Lower Bounds

The optimization of the Evidence Lower Bound (ELBO) serves as the foundational mechanism for tractable variational inference in Bayesian neural networks, circumventing the intractability of computing the exact marginal likelihood. Since the true posterior distribution is generally inaccessible, the inference process is reformulated as minimizing the Kullback-Leibler divergence between an approximate variational distribution and the true posterior [4]. This minimization is mathematically equivalent to maximizing the ELBO, which de-

composes the objective into a likelihood term that encourages data fidelity and a complexity cost represented by the KL divergence against the prior [14]. This duality ensures that the optimization balances fitting the observed data with maintaining parameter simplicity, effectively tightening the lower bound on the log-evidence as the approximation improves.

$$\nabla_{\theta} \log p(\mathbf{y}|\boldsymbol{\theta}) = \sum_{\{i: y_i=1\}} \nabla_{\theta} f_{NN}(\mathbf{x}_i, \boldsymbol{\theta}) - \sum_i g[f_{NN}(\mathbf{x}_i, \boldsymbol{\theta})] \nabla_{\theta} f_{NN}(\mathbf{x}_i, \boldsymbol{\theta}),$$

Figure 15: Chart from 'Bayesian Neural Network Versus Ex-Post Calibration For Prediction Uncertainty' (arXiv:2209.14594)

Practical implementation of this objective often requires addressing the high variance inherent in gradient estimators derived from Monte Carlo sampling. To facilitate efficient backpropagation through stochastic nodes, the reparameterization trick is frequently employed, expressing random variables as deterministic transformations of auxiliary noise distributions. This approach allows for the direct optimization of variational parameters using stochastic gradient descent, transforming the integral formulation of the ELBO into a differentiable loss function. Furthermore, advanced techniques utilize auxiliary distributions within augmented probability spaces to construct tighter bounds on entropy, thereby enhancing the flexibility of the approximate posterior to capture complex, multi-modal dependencies that standard mean-field assumptions might miss.

Despite its computational advantages, the optimization landscape of the ELBO presents challenges regarding the rigidity of restrictive variational families, which can severely limit the expressiveness of the resulting posterior approximation. While standard approaches often assume factorized Gaussian distributions to maintain analytical tractability, such constraints may lead to underestimation of uncertainty, particularly in data-scarce regions or during out-of-distribution detection. Consequently, recent methodological advancements focus on developing more flexible variational families and hybrid training strategies that disentangle aleatoric and epistemic uncertainties. These refinements aim to improve the tightness of the bound without sacrificing the stability required for converging to reliable uncertainty es-

timates in deep learning architectures [15].

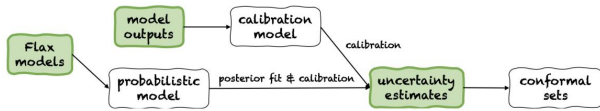


Figure 16: Chart from 'Fortuna: A Library for Uncertainty Quantification in Deep Learning' (arXiv:2302.04019)

3.2 Architectural Integrations of Probabilistic Layers

3.2.1 Probabilistic Formulations in Convolutional Networks

Probabilistic formulations in convolutional networks fundamentally shift the paradigm from learning deterministic point estimates of weights to inferring full posterior distributions over network parameters. By treating weights as random variables governed by prior distributions, these models leverage Bayes' theorem to update beliefs based on observed data, thereby capturing epistemic uncertainty inherent in the model structure. This approach contrasts sharply with standard backpropagation, which yields a single optimal configuration, and instead provides a distributional representation that naturally regularizes the network against overfitting, particularly in data-scarce regimes common in computer vision tasks.

To make such Bayesian inference computationally tractable within deep convolutional architectures, variational inference has emerged as the dominant approximation strategy [9]. Given the intractability of computing the exact posterior, these methods optimize a variational distribution to minimize the Kullback-Leibler divergence from the true posterior, effectively maximizing the evidence lower bound [4]. Techniques such as Bayes by Backpropagation and the reparameterization trick enable efficient gradient-based optimization of these distributions, allowing convolutional layers to learn mean and variance parameters for their filters. Furthermore, innovations like Flipout and multiplicative normalizing flows have been introduced to reduce the variance of gradient estimators and enhance the expressiveness of the approximate posterior, ensuring scalable training without sacrificing the richness of the probabilistic representation.

The integration of probabilistic mechanisms into convolutional operations also facilitates the

disentanglement of aleatoric and epistemic uncertainties during inference. By maintaining stochasticity through methods like Monte Carlo Dropout or explicit weight sampling, the network generates predictive distributions rather than single-point outputs, offering calibrated confidence estimates crucial for safety-critical applications. Recent advancements suggest that even partial stochasticity, where only specific layers are treated probabilistically while others remain deterministic, can offer an optimal trade-off between computational efficiency and uncertainty quantification performance. This hybrid approach preserves the representational power of deep convolutional networks while endowing them with the robustness and interpretability characteristic of Bayesian frameworks.

3.2.2 Uncertainty Propagation in Recurrent and Graph Structures

Propagating uncertainty through recurrent and graph-structured architectures presents distinct challenges compared to feedforward networks due to their iterative nature and complex dependency topologies. In recurrent neural networks, uncertainty estimates must be maintained across temporal steps, where naive propagation often leads to rapidly collapsing or exploding variance, rendering long-term predictions unreliable. Similarly, in graph neural networks, the message-passing mechanism aggregates uncertain features from neighboring nodes, necessitating rigorous handling of correlated errors that arise from shared structural dependencies and varying node degrees.

To address these issues, recent methodologies extend variational inference and Monte Carlo dropout techniques specifically tailored for sequential and relational data. For recurrent structures, approaches often involve maintaining a distribution over hidden states rather than point estimates, utilizing reparameterization tricks to sample latent trajectories while preserving gradient flow through time. In graph domains, uncertainty propagation is frequently modeled by treating node embeddings as random variables, where the aggregation function computes the moments of the resulting posterior distribution. This requires approximating the joint distribution of neighboring nodes to account for the non-independent nature of graph-structured data, often leveraging local linearization or moment-matching techniques to ensure computational tractability.

Despite these advancements, significant limitations remain regarding the disentanglement of aleatoric and epistemic components within these dynamic systems. The recursive application of uncertainty operators in recurrent models can introduce pathological behaviors, such as overconfidence in out-of-distribution sequences, while graph-based methods struggle with scalability when computing full covariance matrices for large-scale networks. Furthermore, the assumption of factorized posteriors over weights or hidden states often fails to capture the complex correlations inherent in temporal dynamics and graph topology. Consequently, current research focuses on developing more robust approximation schemes that balance analytical tractability with the fidelity required for reliable uncertainty quantification in structured deep learning scenarios [15].

3.3 Theoretical Foundations and Limitations

3.3.1 Posterior Consistency and Convergence Properties

Posterior consistency in Bayesian neural networks dictates that as the volume of observed data increases, the approximate posterior distribution must concentrate around the true data-generating parameters. Theoretically, this implies that the predictive uncertainty, specifically the epistemic component, should asymptotically vanish, leaving only the irreducible aleatoric noise. In the infinite data limit, the variance of the posterior predictive distribution converges to the inherent noise floor, ensuring that the model’s confidence aligns with the underlying stochasticity of the target function rather than parameter ambiguity.

However, achieving this convergence in practice is often hindered by the restrictive assumptions inherent in variational inference. Common mean-field approximations, which assume independence among weights, frequently fail to capture the complex correlations within the true posterior, leading to suboptimal convergence rates or inconsistent estimates. While sampling-based methods like Markov Chain Monte Carlo offer theoretical guarantees of consistency, their computational intractability for large-scale networks necessitates approximate methods that may sacrifice asymptotic correctness for scalability, resulting in posteriors that remain overly diffuse even with abundant data.

To mitigate these rigidities, recent advance-

ments explore more expressive posterior families, such as mixtures of Gaussians or transformations designed to capture heavy-tailed distributions. These approaches aim to reduce the Kullback-Leibler divergence gap between the variational approximation and the true posterior, thereby improving the fidelity of uncertainty quantification during extrapolation. Despite these improvements, ensuring that the optimization landscape allows the approximate posterior to consistently track the true posterior as dataset dimensions grow remains a critical challenge, particularly when balancing the trade-off between approximation flexibility and computational efficiency.

3.3.2 Challenges in High-Dimensional Parameter Spaces

The primary obstacle in high-dimensional parameter spaces is the computational intractability of exact Bayesian inference, driven by the exponential growth of the search volume and the complexity of integrating over non-Gaussian posterior distributions. As model architectures deepen, the dimensionality of the weight space expands, rendering standard Markov Chain Monte Carlo methods prohibitively expensive due to slow mixing rates and the curse of dimensionality. This complexity is further exacerbated by the nonlinear interactions between parameters, which create rugged loss landscapes characterized by numerous local minima and saddle points that hinder convergence to global optima.

Consequently, approximate inference techniques such as variational inference and sparse approximations become necessary to reduce computational complexity, yet they introduce their own set of limitations regarding accuracy and calibration. In high-dimensional settings, the assumption of factorized posterior distributions often fails to capture intricate correlations between parameters, leading to underestimation of uncertainty and overconfident predictions. Furthermore, the optimization of variational parameters itself becomes a high-dimensional non-convex problem, where gradient-based methods may struggle to navigate the parameter space effectively without careful initialization or advanced regularization strategies to prevent overfitting.

The challenge is compounded by the sensitivity of model performance to hyperparameter configurations, which scale poorly as the number of layers and units increases. Small perturbations in

high-dimensional spaces can lead to significant deviations in predictive performance, making robust model selection and tuning increasingly difficult. Additionally, the presence of many-to-one mappings in complex physical systems implies that distinct parameter combinations can yield identical outputs, creating identifiability issues that obscure the true underlying data generating process and limit the interpretability of learned representations in deep probabilistic models.

4 Advanced Posterior Approximation and Sampling Techniques

4.1 Deterministic Approximation Methods

4.1.1 Laplace Approximations and Curvature Estimates

The Laplace approximation serves as a foundational technique for Bayesian inference in deep neural networks by approximating the intractable posterior distribution with a Gaussian centered at a local mode of the likelihood. This method relies on a second-order Taylor expansion of the log-posterior around the maximum a posteriori (MAP) estimate, effectively linearizing the network function in the parameter space. The resulting covariance matrix is defined by the inverse of the Hessian of the negative log-likelihood, which encodes the local curvature of the loss landscape. While theoretically elegant, the direct application of this approach to modern deep architectures is often prohibitive due to the immense computational cost and memory requirements associated with storing and inverting the full Hessian matrix.

To address these scalability challenges, recent advancements have focused on efficient curvature estimates that approximate the Hessian without explicit construction. Common strategies include utilizing the Generalized Gauss-Newton matrix, which ensures positive semi-definiteness by approximating the Hessian as the product of the Jacobian transpose and itself, thereby neglecting second-order derivative terms of the network outputs. Further structural assumptions, such as diagonal or low-rank approximations, significantly reduce the parameter complexity from quadratic to linear or near-linear order. These structured approximations enable the practical deployment of Laplace methods in over-parameterized models, allowing for the estimation of epistemic uncertainty through the propagation of parameter variance via the linearized model.

Despite the inherent limitation of assuming a unimodal posterior, which may fail to capture complex multi-modal structures in the weight space, scalable Laplace approximations have demonstrated competitive performance in uncertainty quantification and calibration. The method effectively bridges the gap between point-estimate training and full Bayesian inference by leveraging standard optimization trajectories to identify the expansion point. By integrating these curvature estimates into the predictive distribution, the approach provides a computationally efficient mechanism for distinguishing between aleatoric and epistemic uncertainty, offering a robust alternative to more expensive sampling-based methods like Markov Chain Monte Carlo while maintaining rigorous probabilistic interpretations.

4.1.2 Analytical Propagation of Moments

Analytical propagation of moments offers a deterministic alternative to stochastic sampling by directly computing the statistical properties of network outputs through the layers. Instead of relying on Monte Carlo integration, which can be computationally prohibitive for real-time applications, this approach approximates the posterior distribution of activations by propagating only the first and second moments—mean and covariance—through the network architecture. This method assumes that the pre-activation distributions remain approximately Gaussian or belong to a tractable family, allowing for closed-form updates or efficient numerical approximations at each layer without drawing explicit weight samples.

The core challenge in this framework lies in handling non-linear activation functions, which generally render the exact analytical integration of moments intractable. To address this, techniques such as the probit approximation or linearization around the mean are employed to estimate the transformed mean and variance. For instance, when passing a Gaussian distributed input through a sigmoid or ReLU function, specific closed-form approximations or lookup tables are utilized to map the input moments to the output moments. This ensures that the uncertainty, represented by the diagonal and off-diagonal elements of the covariance matrix, evolves consistently, capturing both the variance of individual units and the inter-dependencies between them.

By maintaining a full or diagonal covariance structure throughout the forward pass, analytical

moment propagation provides a granular view of epistemic and aleatoric uncertainty accumulation. The trace of the resulting covariance matrix reflects the total variance of the predictions, while off-diagonal components reveal correlations between different output classes or features, enhancing interpretability. Although this approach often relies on mean-field assumptions that may ignore complex posterior dependencies, it significantly reduces computational overhead compared to ensemble methods or deep Monte Carlo dropout, making it suitable for scenarios requiring rapid uncertainty estimation with limited computational resources.

4.2 Stochastic Sampling Algorithms

4.2.1 Markov Chain Monte Carlo Dynamics

Markov Chain Monte Carlo (MCMC) dynamics construct a sequence of dependent samples that asymptotically converge to the target Bayesian posterior distribution [6]. Unlike independent sampling techniques, MCMC generates a Markov chain where each state depends probabilistically solely on its predecessor, ensuring that the stationary distribution of the chain matches the intractable posterior. This iterative process allows for the approximation of complex integrals required for Bayesian inference in neural networks, although the inherent dependence between successive samples introduces autocorrelation that can inflate the variance of estimators and slow down convergence.

To mitigate the inefficient random-walk behavior characteristic of basic Metropolis-Hastings algorithms, Hamiltonian Monte Carlo (HMC) incorporates first-order gradient information to guide the sampling trajectory. By introducing an auxiliary momentum vector and simulating Hamiltonian dynamics, HMC proposes distant states with high acceptance probabilities, effectively suppressing random walks and exploring the parameter space more efficiently. However, despite these improvements, standard HMC implementations still struggle with high computational costs per iteration and residual autocorrelations, particularly when applied to high-dimensional neural network weight spaces where calculating full gradients is prohibitive.

Recent advancements have focused on scaling these dynamics to large datasets through stochastic approximations, such as Stochastic Gradient

Langevin Dynamics (SGLD) and Stochastic Gradient Hamiltonian Monte Carlo. These methods replace full-batch gradients with noisy mini-batch estimates, injecting controlled noise to maintain the correct invariant distribution while significantly reducing computational overhead. While these stochastic variants bridge the gap between scalable optimization and rigorous Bayesian sampling, they require careful tuning of step sizes and noise schedules to balance the trade-off between sampling accuracy and the bias introduced by discretization errors in the dynamic simulation.

4.2.2 Langevin Dynamics and Hybrid Sampling Schemes

Stochastic Gradient Langevin Dynamics (SGLD) addresses the scalability limitations of traditional Markov Chain Monte Carlo methods by integrating stochastic gradient estimates directly into the Langevin diffusion process. By injecting controlled Gaussian noise into the parameter updates, SGLD transforms standard optimization trajectories into a Markov chain that asymptotically converges to the true posterior distribution without requiring a Metropolis-Hastings acceptance step. This approach effectively leverages mini-batch computations to approximate the gradient of the log-posterior, making it computationally feasible for large-scale Bayesian neural networks where full-batch gradients are prohibitive.

However, the introduction of stochastic noise from minibatching disrupts the precise Hamiltonian dynamics utilized in methods like Hamiltonian Monte Carlo, often leading to high rejection rates or trajectory deviations in hybrid schemes. To mitigate these issues, advanced hybrid sampling algorithms incorporate friction terms and second-order dynamics, drawing inspiration from statistical mechanical systems to counterbalance stochastic forces. These modifications aim to suppress random-walk behavior and reduce autocorrelation among successive samples, thereby improving the efficiency of exploring complex, multimodal posterior landscapes that variational inference often fails to capture accurately.

Despite these theoretical advantages, practical implementation requires careful tuning of step size schedules and noise variance to ensure convergence while maintaining computational efficiency. The interplay between the diminishing step size and the injected noise dictates the trade-off between exploration capability and the fidelity of the

stationary distribution. Consequently, recent developments focus on adaptive mechanisms that dynamically adjust these hyperparameters, allowing hybrid schemes to robustly approximate posterior uncertainties in high-dimensional spaces while retaining the scalability benefits inherent to stochastic gradient-based approaches.

4.3 Ensemble-Based Posterior Representations

4.3.1 Deep Ensembles as Bayesian Model Averaging

Deep ensembles function as a pragmatic approximation of Bayesian Model Averaging (BMA) by aggregating predictions from multiple independent neural networks trained with distinct random initializations. Unlike traditional Bayesian approaches that explicitly define a prior and compute a posterior distribution over weights, this method relies on the diversity induced by non-convex optimization landscapes to generate a set of disparate point estimates. By minimizing the negative log-likelihood for each member independently, the ensemble effectively samples from different modes of the loss surface, creating a mixture distribution that captures epistemic uncertainty through predictive variance.

Although deep ensembles lack a formal probabilistic derivation where the posterior is analytically computed, they empirically replicate the benefits of BMA by averaging the predictive distributions of constituent models. This approach yields well-calibrated error estimates and robust performance, particularly in scenarios where the true posterior deviates significantly from simple parametric assumptions like Bernoulli or Gaussian distributions. The method’s strength lies in its ability to explore the parameter space more effectively than single-point estimators, as each network converges to a unique solution that contributes to a broader coverage of plausible functions.

From a computational perspective, deep ensembles incur a linear increase in training time and memory proportional to the number of ensemble members, yet they often outperform sampling-based methods like Monte Carlo Dropout in terms of predictive accuracy and uncertainty calibration [12]. While critics argue that the technique is not strictly Bayesian due to the absence of an explicit prior, recent theoretical analyses suggest that the implicit bias of stochastic gradient descent com-

bined with random initialization serves as an effective surrogate for Bayesian inference. Consequently, deep ensembles have established themselves as a state-of-the-art baseline for uncertainty quantification, bridging the gap between frequentist ensemble learning and Bayesian model averaging [12].

4.3.2 Functional Space Approximations via Ensembling

Functional space approximations via ensembling offer a pragmatic approach to Bayesian inference by constructing a mixture distribution from multiple independent variational approximations. Instead of relying on a single parametric family, such as a mean-field Gaussian, this strategy learns an ensemble of distinct posterior estimates. The resulting aggregate distribution effectively forms a Gaussian mixture model, which possesses the capacity to capture multi-modal structures inherent in the true posterior that unimodal approximations inevitably miss.

While this ensemble-based mixture cannot perfectly replicate the intricate geometric complexities of the exact posterior, it provides a significantly more accurate representation of the marginal distribution over latent variables compared to standard variational methods. Empirical analyses indicate that the marginals generated through ensembling closely align with those derived from computationally expensive Markov Chain Monte Carlo samplers. This alignment demonstrates that combining simple variational families can yield high-fidelity approximations of the predictive distribution without requiring explicit modeling of complex dependency structures within a single network.

A critical advantage of this methodology lies in its computational efficiency during the inference phase. Despite the increased expressiveness afforded by the mixture model, the time required to compute marginal predictions remains comparable to that of a single Gaussian variational approximation. This scalability makes ensembling a highly effective tool for approximate Bayesian deep learning, bridging the gap between the tractability of variational inference and the expressive power needed for robust uncertainty quantification in high-dimensional functional spaces [1].

5 Future Directions

Despite significant advancements in Bayesian Deep Learning, current methodologies exhibit persistent limitations regarding the trade-off between computational scalability and theoretical fidelity. While sampling-based approaches like Markov Chain Monte Carlo offer asymptotic exactness, they remain prohibitively expensive for high-dimensional architectures and real-time applications. Conversely, scalable variational inference techniques often rely on restrictive mean-field assumptions that fail to capture complex, multi-modal posterior structures, leading to systematic underestimation of uncertainty. Furthermore, existing calibration mechanisms, particularly post-hoc adjustments, demonstrate fragility under severe distribution shifts and non-stationary environments, as static mapping functions cannot adapt to dynamic changes in data characteristics. There is also a notable gap in effectively disentangling aleatoric and epistemic uncertainty in scenarios involving heavy-tailed noise or significant model misspecification, limiting the reliability of these systems in safety-critical domains where distinguishing reducible ignorance from irreducible data noise is paramount.

Future research must prioritize the development of expressive yet computationally efficient inference algorithms that transcend current variational constraints. Investigating structured variational families, such as those leveraging normalizing flows or hierarchical priors, could enable the accurate approximation of complex posterior geometries without sacrificing the scalability required for modern deep networks. Simultaneously, there is a critical need for adaptive calibration frameworks that operate intrinsically within the learning process and remain robust to distribution shifts. This involves designing dynamic calibration objectives that evolve alongside the model during training, potentially integrating meta-learning strategies to adjust uncertainty estimates in response to changing environmental conditions. Additionally, future work should focus on creating unified evaluation benchmarks that rigorously test uncertainty quantification under extreme out-of-distribution scenarios, moving beyond standard accuracy metrics to assess the stability of predictive intervals and the fidelity of uncertainty decomposition across diverse operational contexts.

Addressing these challenges will fundamentally

transform the deployment of artificial intelligence in high-stakes sectors such as autonomous driving, healthcare, and financial risk management. By establishing models that can reliably articulate their own limitations and distinguish between different sources of uncertainty, the proposed research directions will foster a new generation of trustworthy AI systems capable of safe decision-making under ambiguity. Enhanced robustness against distribution shifts and improved calibration will mitigate the risks associated with overconfident predictions, thereby facilitating regulatory approval and broader societal acceptance of automated decision-support tools. Ultimately, bridging the gap between theoretical probabilistic rigor and practical scalability will ensure that deep learning systems not only achieve high performance but also maintain verifiable reliability when facing the unpredictable complexities of the real world.

6 Conclusion

This survey has synthesized the current landscape of Bayesian Deep Learning, emphasizing the critical transition from deterministic accuracy to reliable uncertainty quantification in safety-critical applications. We systematically evaluated the trade-offs between stochastic inference methods, noting that while Markov Chain Monte Carlo techniques offer theoretical exactness, Variational Inference provides the necessary scalability for modern high-dimensional architectures, albeit often at the cost of underestimating posterior variance. Our analysis of evaluation frameworks highlighted the insufficiency of standard accuracy metrics, advocating instead for proper scoring rules, reliability diagrams, and coverage rates to rigorously assess predictive distribution quality. Furthermore, we contrasted post-hoc calibration mechanisms, which efficiently correct overconfidence in static environments, with intrinsic calibration strategies that embed uncertainty awareness directly into the learning objective. A central finding remains the vital distinction between aleatoric and epistemic uncertainty; understanding this decomposition is essential for deploying models that can differentiate between irreducible data noise and reducible model ignorance, thereby enabling more informed decision-making under distribution shifts.

The significance of this work lies in its establishment of a unified taxonomy that bridges the gap between abstract probabilistic theory and em-

pirical deep learning practice. By critically analyzing the interplay between inference algorithms, calibration techniques, and uncertainty decomposition, this survey provides a cohesive framework for researchers and practitioners to navigate the complex design space of trustworthy AI. Unlike previous reviews limited to specific algorithms or domains, this comprehensive examination elucidates how theoretical fidelity impacts computational practicality, offering clear guidance on selecting modeling choices based on operational constraints. The integration of diverse evaluation metrics and the detailed comparison of approximation strategies serve as a foundational reference for diagnosing miscalibration and enhancing the robustness of neural networks. Ultimately, this synthesis fosters a deeper understanding of how to construct systems that not only predict accurately but also articulately express their own limitations, a prerequisite for deploying artificial intelligence in high-stakes scenarios such as autonomous driving and medical diagnostics.

Looking forward, the field must address persistent challenges regarding the robustness of uncertainty estimates under severe model misspecification and non-stationary environments. Future research should prioritize the development of more expressive variational families that can capture complex multi-modal posterior structures without sacrificing computational efficiency, moving beyond restrictive mean-field assumptions. There is an urgent need for dynamic calibration mechanisms that adapt to shifting data distributions in real-time, rather than relying on static post-hoc adjustments derived from held-out datasets. We call upon the community to adopt standardized, comprehensive evaluation protocols that simultaneously assess calibration, sharpness, and coverage to ensure genuine reliability. By focusing on these frontiers, researchers can develop next-generation Bayesian deep learning systems that are not only theoretically sound but also practically resilient, ultimately realizing the vision of artificial intelligence that is both powerful and profoundly trustworthy.

References

[1] LEARNING ACTIVE SUBSPACES FOR EFFECTIVE AND SCALABLE UNCERTAINTY QUANTIFICATION IN DEEP NEURAL NETWORKS

[2] Bayesian Optimized Deep Ensemble for Uncertainty Quantification of Deep Neural Networks a System Safety Case Study on Sodium Fast Reactor Thermal Stratification Modeling

[3] Comparing the quality of neural network uncertainty estimates for classification problems

[4] PARAMETERS ESTIMATION FROM THE 21 CM SIGNAL USING VARIATIONAL INFERENCE

[5] On the Uncertain Single-View Depths in Colonoscopies

[6] Quantification of Predictive Uncertainty via Inference-Time Sampling

[7] Bayesian Learning for Neural Networks an algorithmic survey

[8] From Aleatoric to Epistemic Exploring Uncertainty Quantification Techniques in Artificial Intelligence

[9] BI-RADS prediction of mammographic masses using uncertainty information extracted from a Bayesian Deep Learning model

[10] Bayesian Neural Networks for Uncertainty Estimation of Imaging Biomarkers

[11] Estimating Uncertainty and Interpretability in Deep Learning for Coronavirus (COVID-19) Detection

[12] Constraint-Aware Neurosymbolic Uncertainty Quantification with Bayesian Deep Learning for Scientific Discovery

[13] BORCH A DEEP UNIVERSAL PROBABILISTIC PROGRAMMING LANGUAGE

[14] Bayesian Neural Network Versus Ex-Post Calibration For Prediction Uncertainty

[15] Fortuna A Library for Uncertainty Quantification in Deep Learning